

Lecture 05(a): Strategic Patrolling and Leader-Follower Games

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Key Preferences

- Bernhard Von Stengel and Shmuel Zamir. “Leadership games with convex strategy sets”. In: *Games and Economic Behavior* 69.2 (2010), pp. 446–457
- James Pita et al. “Deployed ARMOR protection: the application of a game theoretic model for security at the Los Angeles International Airport”. In: 2008
- Vincent Conitzer and Tuomas Sandholm. “Computing the optimal strategy to commit to”. In: *Proceedings of the 7th ACM conference on Electronic commerce*. 2006, pp. 82–90

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Stackelberg Security Games (SSGs)

As one of the key successful game theory applications, **Stackelberg Security Games (SSGs)** have proven effective in optimising resource allocation by modelling defender-attacker interactions.

Key Applications:

- **Airport Security:** Protecting eight terminals at Los Angeles International Airport [Pit+08].
- **Conservation:** Safeguarding biodiversity across 2,500+ square kilometers [FST15].
- **Passenger Screening:** Managing security for 800M annual travelers in the USA [Bro+16].

Stackelberg equilibrium[VZ10] models first-mover advantage in games with a Leader and a Follower.

Stackelberg Game Example

- The leader commits to a strategy first, and the follower optimises their reward based on the leader's choice.

L/F	C	D
A	2, 1	4, 0
B	1, 0	3, 2

- Nash Equilibrium: Leader plays A , follower plays C , payoff = 2 for the leader.
- Stackelberg Advantage: Leader commits to B , follower plays D , payoff = 3 for the leader.
- Mixed Strategy: Leader plays A and B with probability 0.5, follower plays D , payoff = 3.5 for the leader.

Nash vs. Stackelberg Equilibrium in Inspection Games

- Inspector decides whether to inspect; inspectee chooses whether to cheat [Kro22].
- The inspector aims to deter cheating, incurring a cost of -1 for inspecting and a heavy penalty if cheating occurs, regardless of detection.

L/F	Cheat	No Cheat
Inspect	$(-6, -9)$	$(-1, 0)$
No Inspect	$(-10, 1)$	$(0, 0)$

- A single unique **Nash Equilibrium**:
 - Inspector inspects with $p = \frac{1}{10}$; inspectee cheats with $q = \frac{1}{5}$.
 - Expected utilities: $(-2, 0)$.
- **Stronge Stackelberg Equilibrium**: F resolves ties in L 's favor.
 - Inspector inspects with $p = \frac{1}{10}$, deterring cheating ($q = 0$).
 - Expected utilities: $(-\frac{1}{10}, 0)$, better for the inspector.
- **Weak Stackelberg Equilibrium**: ties are broken adversarially
 - Requires $p > \frac{1}{10}$; does not exist as the inspector can always improve by adjusting p .

Stackelberg Equilibrium in Finite Games

Definition: In a two-player finite game, let:

- Σ_1 and Σ_2 denote the mixed strategy sets for Player 1 (Leader) and Player 2 (Follower).
- $u_1(\sigma_1, \sigma_2)$ and $u_2(\sigma_1, \sigma_2)$ represent the utilities for Players 1 and 2, given mixed strategies $\sigma_1 \in \Sigma_1$ and $\sigma_2 \in \Sigma_2$.

Player 2's best response set to σ_1 is:

$$BR_2(\sigma_1) = \{\sigma_2 \in \Sigma_2 : u_2(\sigma_1, \sigma_2) \geq u_2(\sigma_1, \sigma'_2), \forall \sigma'_2 \in \Sigma_2\}.$$

$\sigma_1^* \in \Sigma_1$ is **SSE** if: $\sigma_1^* = \arg \max_{\sigma_1 \in \Sigma_1} \max_{\sigma_2 \in BR_2(\sigma_1)} u_1(\sigma_1, \sigma_2)$, where ties in $BR_2(\sigma_1)$ favor Player 1.

In addition, $\sigma_1^* \in \Sigma_1$ is a **Weak Stackelberg Equilibrium (WSE)** if:

$$\sigma_1^* = \arg \sup_{\sigma_1 \in \Sigma_1} \min_{\sigma_2 \in BR_2(\sigma_1)} u_1(\sigma_1, \sigma_2),$$

with ties are broken adversarially [VZ10].

LP Formulation for Pure Best Strategy Response Case¹

If we assume Follower (Player 2) commits to a best pure strategy response $a_2 \in BR_2(\sigma_1)$, we compute the **Strong Stackelberg Equilibrium (SSE)** by:

$$\max_{\sigma_1 \in \Sigma_1} \sum_{a_1 \in A_1} \sigma_1(a_1) u_1(a_1, a_2),$$

subject to:

$$\sum_{a_1 \in A_1} \sigma_1(a_1) u_2(a_1, a_2) \geq \sum_{a_1 \in A_1} \sigma_1(a_1) u_2(a_1, a'_2), \quad \forall a'_2 \in A_2.$$

Algorithm:

- Iterate over all $a_2 \in A_2$.
- Solve the LP for each $a_2 \in A_2$ and select the solution σ_2^* with the highest value. For general solutions, refer to [CS06]

¹Christian Kroer. *Lecture Note 16: Stackelberg Equilibrium and Security Games*. http://www.columbia.edu/~ck2945/files/ai_games_markets/lecture_note_16_stackelberg_games.pdf. Accessed: 2025-01-19. 2022.

Theorem: Stackelberg vs. Nash Equilibrium

Theorem:

In general-sum, Leader achieves weakly greater utility in a Strong Stackelberg Equilibrium (SSE) than in any Nash equilibrium.

Proof

- Let (σ_L, σ_F) be the Nash equilibrium yielding the leader's highest utility.
- In SSE, the follower breaks ties in favor of the leader:
 - If the leader commits to σ_L , the follower selects $\sigma_F \in BR(\sigma_L)$, yielding the same utility as the Nash equilibrium, or
 - Chooses $\sigma'_F \in BR(\sigma_L)$, which provides higher utility for the leader.



The leader's payoff in WSE is at least as high as in some Nash equilibrium [VZ10], with Stackelberg payoffs $[L, H]$ strictly above Nash. However, in zero-sum games, Stackelberg equilibrium coincides with Nash equilibrium.

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References I

- [Bro+16] Matthew Brown et al. “One size does not fit all: A game-theoretic approach for dynamically and effectively screening for threats”. In: *Thirtieth AAAI Conference on Artificial Intelligence*. 2016.
- [CS06] Vincent Conitzer and Tuomas Sandholm. “Computing the optimal strategy to commit to”. In: *Proceedings of the 7th ACM conference on Electronic commerce*. 2006, pp. 82–90.
- [FST15] Fei Fang, Peter Stone, and Milind Tambe. “When security games go green: Designing defender strategies to prevent poaching and illegal fishing”. In: *Twenty-Fourth International Joint Conference on Artificial Intelligence*. 2015.

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- [Kro22] Christian Kroer. *Lecture Note 16: Stackelberg Equilibrium and Security Games*. http://www.columbia.edu/~ck2945/files/ai_games_markets/lecture_note_16_stackelberg_games.pdf. Accessed: 2025-01-19. 2022.
- [Pit+08] James Pita et al. “Deployed ARMOR protection: the application of a game theoretic model for security at the Los Angeles International Airport”. In: 2008.
- [VZ10] Bernhard Von Stengel and Shmuel Zamir. “Leadership games with convex strategy sets”. In: *Games and Economic Behavior* 69.2 (2010), pp. 446–457.